Graph Neural Networks

AI506 — Advanced Machine Learning

Simon Holm

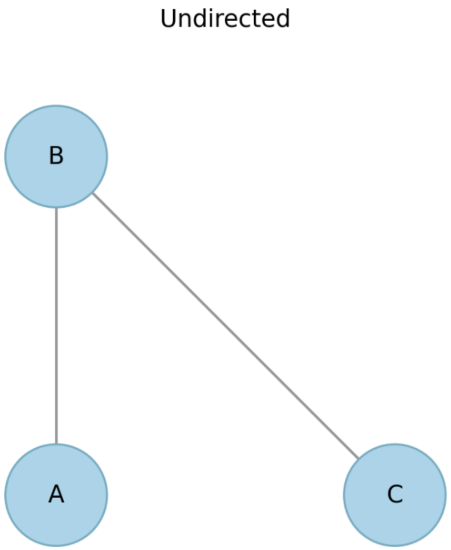
May 2026

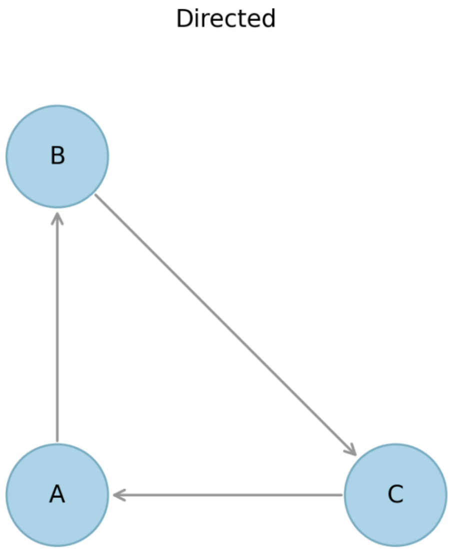
The graph structure

Let $\texttt{nodes} = \{\mathrm{A}, \mathrm{B}, \mathrm{C}\}$

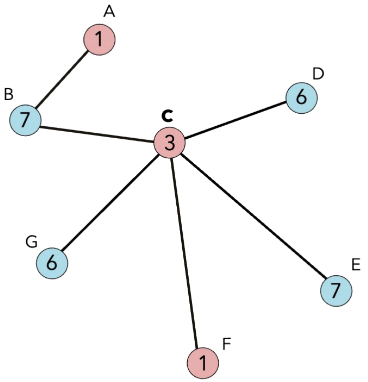
$$\texttt{edges} = \{(A, B), (B, A), (B, C), (C, A)\}$$

$$\texttt{edges} = \{(A, B), (B, C)\}$$





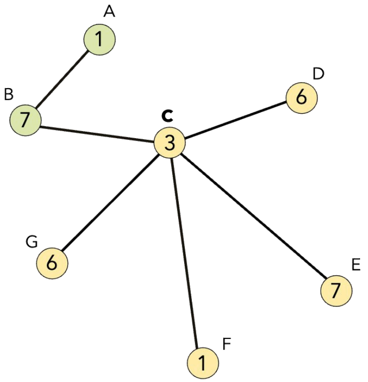
Common GNN tasks



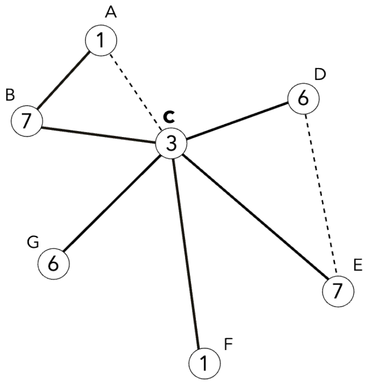
Node Classification

Toxic

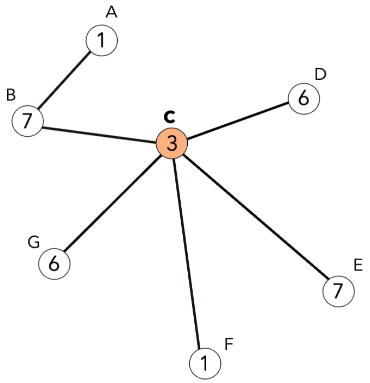
Graph Classification



Node Clustering



Link Prediction



Influence Maximization

Common GNN tasks

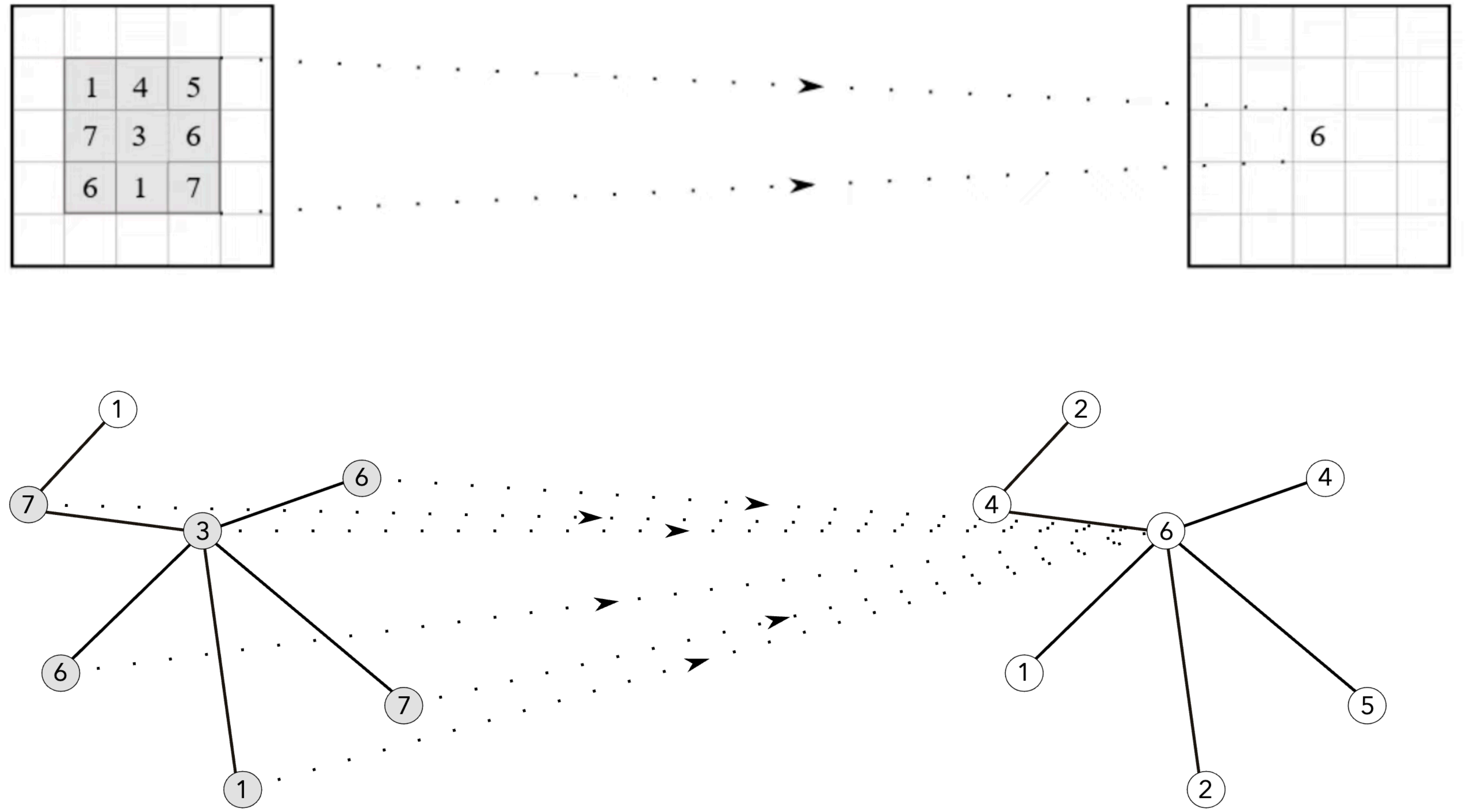
For node classification:

Let X = H^0

$$H^{(l+1)} = \sigma\left(\hat{A} H^{(l)} W^{(k)} + b^{(l)}\right)$$

$$Z = \mathrm{softmax}\left(\hat{A}\ \mathrm{ReLU}\left(z^{(l-1)}\right) W^{(l)}\right)$$

Graph convolution



Message passing framework

$$h_i^{(l+1)} = \sigma\left(\mathrm{UPD}\left(h_i^{(l)}, m_i^{(l)}\right)\right)$$

$$\mathrm{UPD}\left(h_i^{(l)}, m_i^{(l)}\right) = W_h \cdot h_i^{(l)} + W_{\mathcal{N}} m_i^{(l)} + b^{(l)}$$

where

$$m_i^{(l)} = \mathrm{AGG}_{j \in \mathcal{N}(i)} g\left(h_i^{(l)}, h_j^{(l)}\right)$$

$$g = \text{some transformation of } h_i \text{ and } h_j$$

Aggregation

$$\mathrm{AGG}_{\mathrm{sum}} = \sum_{j \in \mathcal{N}(i)} h_j^{(l)}$$

$$\mathrm{AGG}_{\mathrm{avg}} = \frac{1}{\|\mathcal{N}(i)\|} \sum_{j \in \mathcal{N}(i)} h_j^{(l)}$$

$$\mathrm{AGG}_{\mathrm{max}} = \max_{j \in \mathcal{N}(i)} \left(h_j^{(l)} \right)$$

Graph attention (GAT)

- Edge attributes (weight)

$$g\left(h_i^{(l)}, h_j^{(l)}\right) = a_{ij} \cdot h_j^{(l)}$$

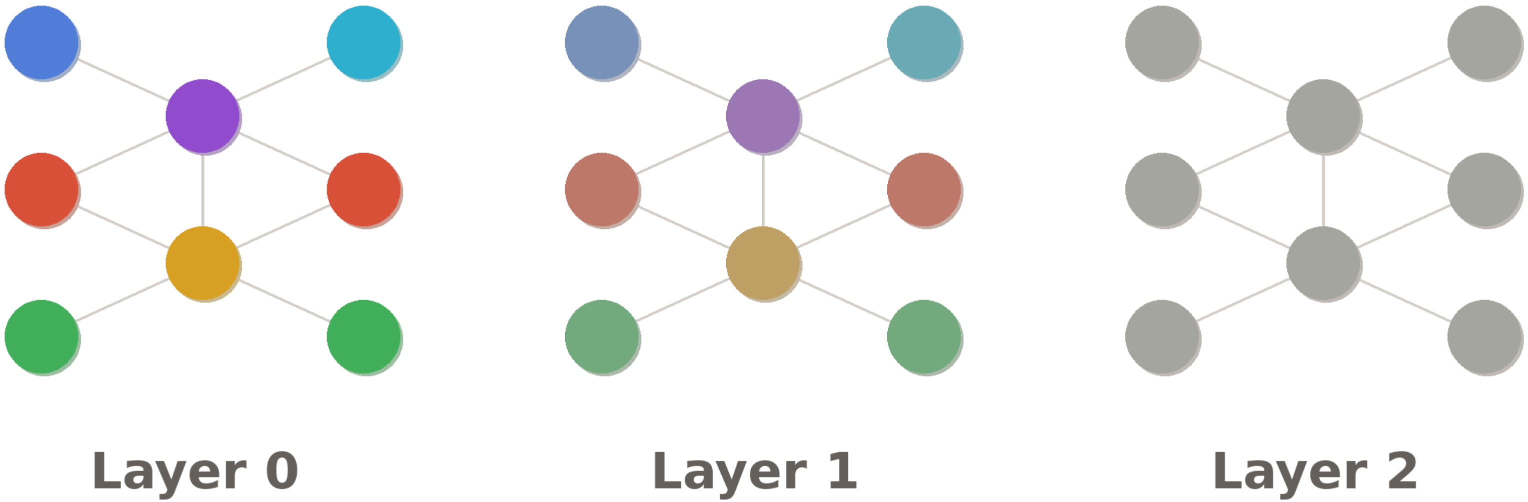
$$\text{where } a_{ij} = \text{softmax}_{\mathcal{N}(i)}\left(a^T \left[W h_i \mathbin{\Vert} W h_j\right]\right)$$

Where a is a learnable vector.

$$\text{AGG}_{\text{sum}} = \sum_{j \in \mathcal{N}(i)} a_{ij} h_j^{(l)}$$

Over-smoothing

$$\mathrm{AGG}_{\mathrm{avg}} = \frac{1}{\|\mathcal{N}(i)\|} \sum_{j \in \mathcal{N}(i)} h_j^{(l+1)} = \frac{1}{\|\mathcal{N}(i)\|} \sum_{j \in \mathcal{N}(i)} \left(\frac{1}{\|\mathcal{N}(j)\|} \sum_{k \in \mathcal{N}(j)} h_k^{(l)} \right)$$



Jumping knowledge

Concatenation:

$$z_i = \mathrm{MLP}\left(h_i^{(1)} \mathbin{+\!\!+} h_i^{(2)} \mathbin{+\!\!+} \dots \mathbin{+\!\!+} h_i^{(l)}\right)$$

Max pooling:

$$z_i = \mathrm{MAX}\left(h_i^{(1)}, h_i^{(2)}, \dots, h_i^{(l)}\right)$$

LSTM:

$$z_i = \mathrm{LSTM}\left(h_i^{(1)}, h_i^{(2)}, \dots, h_i^{(l)}\right)$$